(DEMO) From Natural Language to Live 5G API Calls: A CAPIF-Compliant, Multi-LLM Intent Orchestration Platform

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***Abstract*—We demonstrate an open-source orchestration platform that turns free-text operator intents into live CAMARA API calls whose QoS is authorized by a 3GPP PCF. A two-phase large-language-model (LLM) planner selects the CAMARA service and builds a schema-grounded payload; a three-layer deterministic validator (schema, telecom semantics, live CAPIF registry) gates every call before it reaches a CAPIF-registered provider. The southbound path runs on a containerized Open5GS 5G core with UERANSIM-emulated UEs: a QoD request travels CAMARA → NEF (TS 29.122) → PCF (TS 29.514) and is bound to the UE's PDU session. GPT-4o-mini, Claude Haiku 4.5 and Gemini 2.5 Flash are switchable at runtime. In recorded live runs a request completes in 2.2–4.8 s with GPT-4o-mini or Claude Haiku 4.5 (6.9–8.2 s with Gemini); LLM planning accounts for 78–99 % of it. On 300 unique bilingual intents, all LLM planners reach an Intent-to-API Workflow Success Rate (IWSR) of 95.3–97.3 %, versus 83.3 % for a keyword baseline. Code, testbed, datasets and recorded live runs: https://github.com/doadurak/mini-telco-platform.**

***Index Terms*—CAPIF, CAMARA, intent-based networking, LLM orchestration, 5G API exposure, Open5GS**

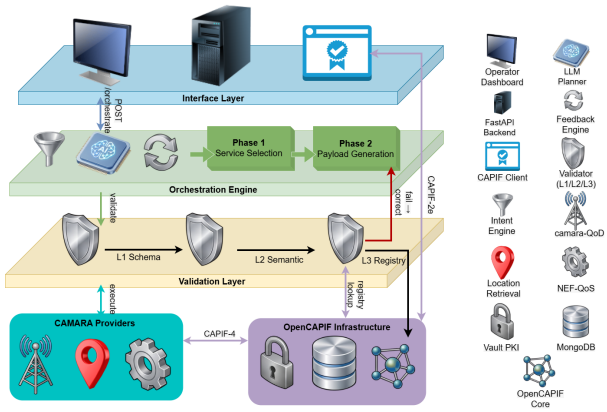


Fig. 1. Platform architecture. The CAPIF client onboards with a Vault-signed certificate; onboarding and discovery use mTLS. Every plan passes L1–L3 before a provider is called. QoS requests reach the Open5GS PCF through NEF-QoS.

I. Introduction

Zero-touch Beyond-5G management needs orchestration layers that translate high-level intents into standards-compliant API invocations [1]. 3GPP CAPIF [2] and GSMA CAMARA [3] standardize how verticals discover and consume network capabilities, but closing the intent-to-API gap on a *live* exposure stack remains open. OpenCAPIF [4] provides exposure infrastructure without NL planning; Turkovic et al. [5] ran live CAMARA QoD with hand-written payloads; Maatouk et al. [6] identify hallucination as the key risk of telecom LLMs, and Liu et al. [7] report 40 % JSON-schema compliance for ungrounded GPT-4. We contribute an open platform that combines live CAPIF onboarding, schema-grounded multi-LLM planning, a deterministic three-layer validator, and CAMARA execution down to a 3GPP PCF, all reproducible from one script.

II. Platform Architecture

Planning. A FastAPI backend exposes `POST /orchestrate` (Fig. 1). Phase 1 of the LLM planner selects one of four CAMARA services (QoD v1.1, QoS Profiles, QoS Provisioning, Location Retrieval v0.5) from a schema-grounded catalog; Phase 2 fills the payload with the operation's OpenAPI schema injected [8]. `POST /config/provider` switches the LLM backend at runtime. Intent-level QoS classes (e.g. `QOS_E_STREAMING`) are mapped to the operator profiles the network enforces (`QOS_E/S/M/L`) by target bit rate.

Validation. L1 checks the Pydantic schema (Schema Compliance Rate, SCR); L2 applies telecom rules: E.164 numbers, profiles mappable to an operator profile, durations in $[1, 7200]$ s, unicast IPv4 (Semantic Validity Rate, SVR). L3 queries the live CAPIF registry to confirm the selected service is published, which catches non-existent services but not a wrong-but-registered one; the latter shows up as an IWSR miss against ground truth. Failures trigger a layer-tagged correction prompt (at most three retries).

Execution. CAPIF onboarding and discovery run over mTLS; the provider call itself is a CAMARA northbound HTTP request inside the testbed. For QoS, the adapter resolves the phone number to the UE's IP from the 5GC subscriber database, the CAMARA QoD provider forwards an AsSes-

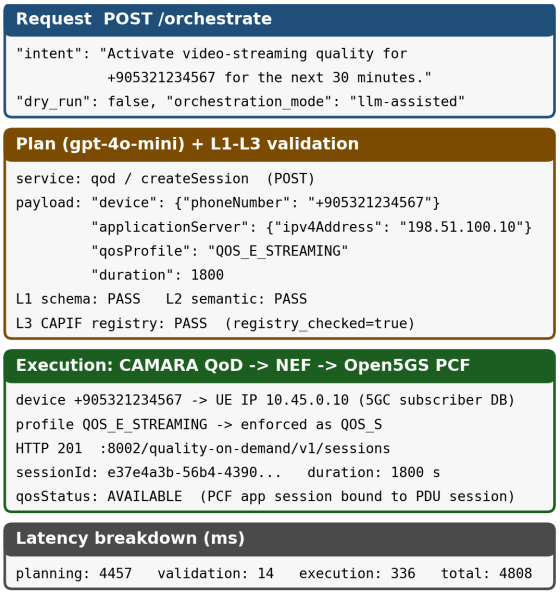


Fig. 2. Recorded live response (S1, GPT-4o-mini, `dry_run:false`): plan, L1–L3 verdicts with `registry_checked=true`, MSISDN-to-UE-IP resolution, profile mapping, the QoD response after the PCF authorized the session, and the per-stage latency.

sionWithQoS subscription to the NEF, and the NEF creates a PCF application session (Npcf_PolicyAuthorization) bound to the UE's PDU session. Location requests are served by the NEF Monitoring Event API from its emulated core data source.

III. Live Demonstration

The full stack (OpenCAPIF, four CAMARA/NEF providers, Open5GS, UERANSIM gNB with three UEs) runs on one 16 GB host via `start.sh`; the only external dependencies are the LLM APIs. Each response shows plan, per-layer verdicts, provider result and per-stage latency; all scenarios run with `dry_run:false` and are versioned in the repository. **S1 (QoD):** *"Activate video-streaming quality for +905321234567 for the next 30 minutes"* yields `QOS_E_STREAMING`, 1800 s, enforced as `QOS_S`; the PCF accepts the application session and QoD returns HTTP 201, `AVAILABLE` (Fig. 2). **S2 (Location):** the NEF returns a WGS-84 area with HTTP 200. **S3 (ambiguity):** dataset entry *"Provide network quality improvement for device +905551112233 for a long time"* (ground truth: QoD). In the benchmark run GPT-4o-mini chose QoD while Claude and Gemini chose QoS Provisioning; in our recorded live run all three chose QoS Provisioning, which the PCF accepted (`ACTIVE`). Every plan passes L1–L3 and executes, showing visitors that the validator guarantees well-formed, registered calls, not intent correctness. **S4 (backend switch):** re-running S1 after each switch yields identical CAMARA payloads from all three LLMs. In the nine runs, planning takes 2.0–4.5 s (GPT, Claude) or 6.4–8.0 s (Gemini), validation 13–27 ms, execution 0.03–0.6 s. UERANSIM does not implement NGAP PDU Session Resource Modify, so the dedicated QoS flow is authorized and signalled by the core but not enforced on the emulated radio.

TABLE I
300-ENTRY OFFLINE EVALUATION (ONE RUN PER INTENT)

Planner	IWSR	SCR	SVR	HR	SS
Keyword baseline	83.3 %	100.0 %	100.0 %	16.7 %	0.962
GPT-4o-mini	**97.3 %**	100.0 %	99.0 %	**2.3 %**	**0.975**
Claude Haiku 4.5	95.3 %	99.3 %	**99.3 %**	4.0 %	0.927
Gemini 2.5 Flash	96.0 %	100.0 %	98.3 %	3.0 %	0.945

IV. Evaluation

The dataset holds 300 unique intents (152 EN / 148 TR; 96 easy, 94 medium, 110 hard) over the four services plus 25 ambiguous edge cases. The authors wrote the ground truth (service, operation, payload checks) from the CAMARA OpenAPI constraints and NEF test fixtures, and supersedes a 60-entry pilot kept in `datasets/legacy_pilot_60/`. IWSR counts plans with the correct service that pass validation. HR is the share of wrong service selections. SS is $1-\sigma$ of IWSR across the three difficulty strata (robustness to difficulty, not run-to-run repeatability). The benchmark makes no provider calls. We re-validated all 1,200 recorded plans against the live CAPIF registry; L3 rejected none, so IWSR is unchanged. Eleven multi-intent prompts first failed because Phase 1 returned a list of plans; after adding multi-step support we re-ran exactly those entries (all now correct, flagged in the result files). The baseline's 50 failures (Table I) are all wrong-service selections on phrasings outside its keyword rules. The 8–14 LLM failures per model are 7–12 wrong-service selections (mostly QoD vs. QoS Provisioning, as in S3) and 1–3 L1/L2 rejections; hard intents account for most of them (hard IWSR 87–95 % vs. 99–100 % for easy and medium).

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